shuffle analysis

Data import and merging

```
peter <- read_csv("Peter_wash_shuffle_data.csv")
sam <- read_csv("Sam_wash_shuffle_data.csv")
caleb <- read_csv("Caleb_wash_shuffle_data_combined.csv")
jordan <- read_csv("Jordan_wash_shuffle_data.csv")
seena <- read_csv("Seena_wash_shuffle_data.csv")

peter_wide <- peter %>%
  pivot_wider(id_cols=c(name, sequence_id, trial_id),
              names_from=start_position,
              values_from=end_position)

sam_wide <- sam %>%
  pivot_wider(id_cols=c(name, sequence_id, trial_id),
              names_from=start_position,
              values_from=end_position)

caleb_wide <- caleb %>%
  pivot_wider(id_cols=c(name, sequence_id, trial_id),
              names_from=start_position,
              values_from=end_position)

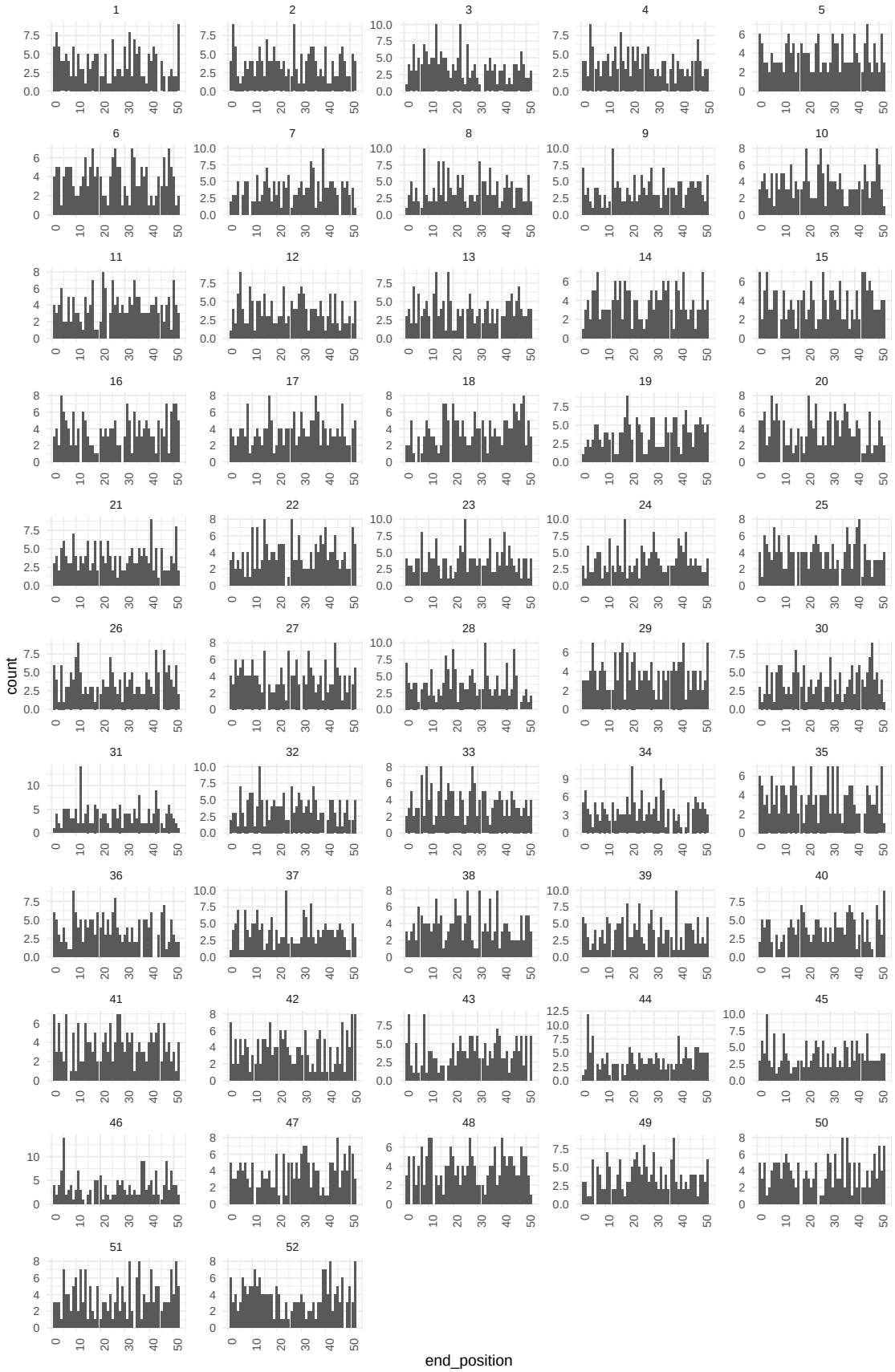
jordan_wide <- jordan %>%
  pivot_wider(id_cols=c(name, sequence_id, trial_id),
              names_from=start_position,
              values_from=end_position)

seena_wide <- seena %>%
  pivot_wider(id_cols=c(name, sequence_id, trial_id),
              names_from=start_position,
              values_from=end_position)
```

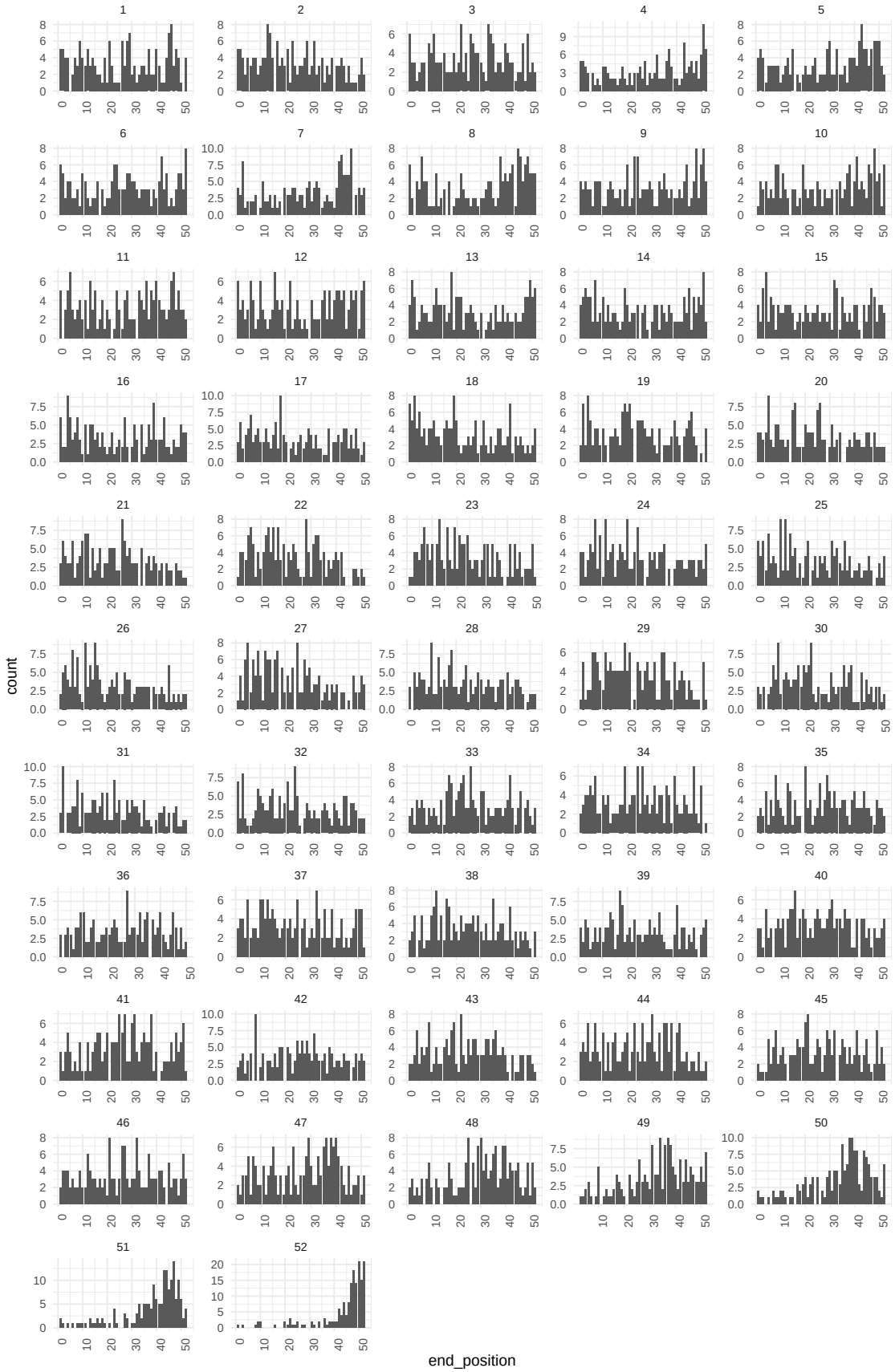
Graphs

Where does the top card go?

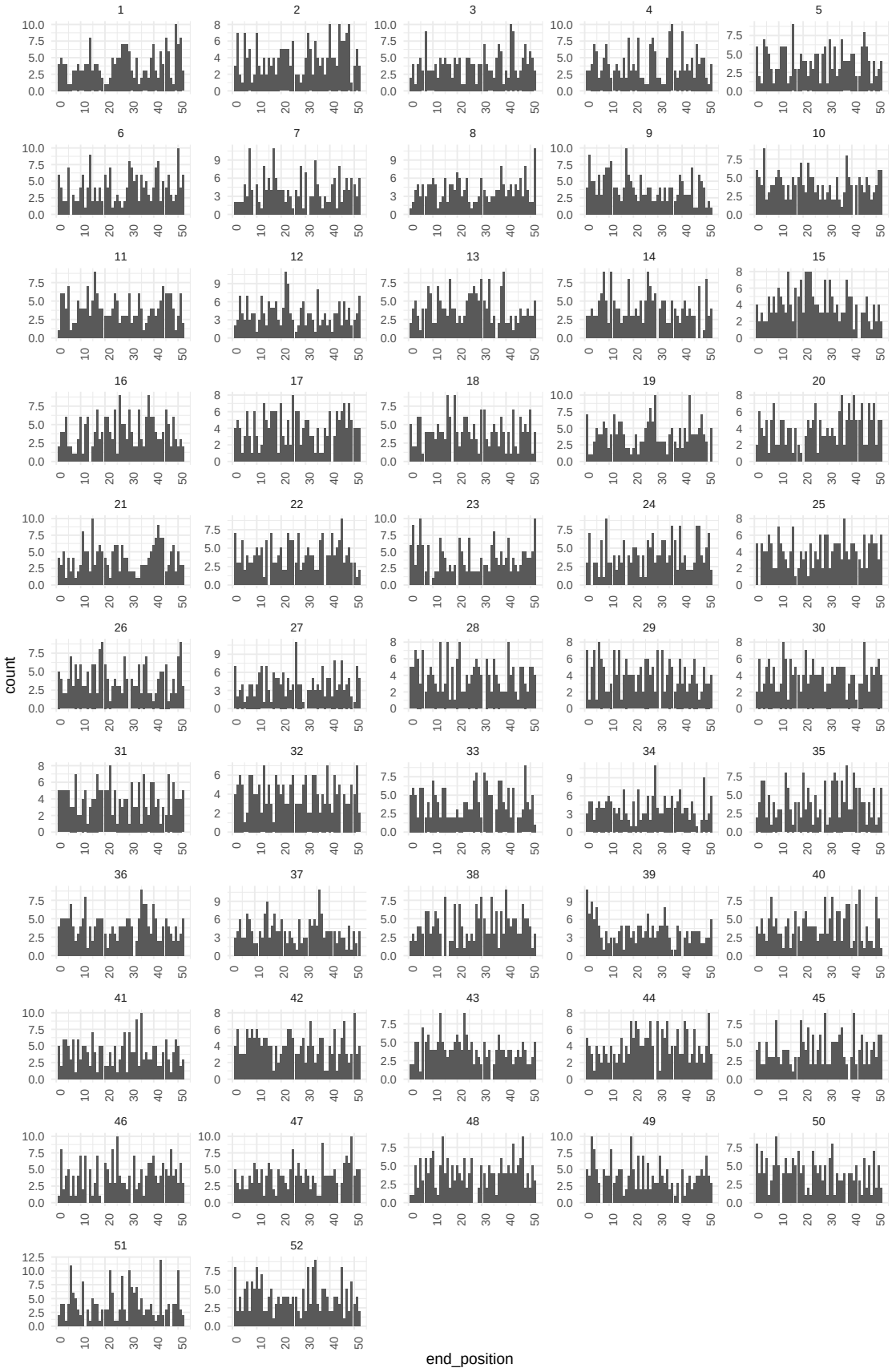
```
ggplot(caleb, aes(x=end_position)) +
  geom_bar() +
  facet_wrap(~ start_position, ncol=5, scales="free") +
  theme_minimal(base_size=8) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```


count
end_position

```r
ggplot(peter, aes(x=end_position)) +
  geom_bar() +
  facet_wrap(~ start_position, ncol=5, scales="free") +
  theme_minimal(base_size=8) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```


count
end_position

```r
ggplot(sam, aes(x=end_position)) +
  geom_bar() +
  facet_wrap(~ start_position, ncol=5, scales="free") +
  theme_minimal(base_size=8) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```


1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
count
end_position

Footrule

Unadjusted footrule calculations

```
peter_fr <- NA
for(i in 1:dim(peter_wide)[1]){
  peter_fr[i] <- Spearman(1:52, as.numeric(as.vector(peter_wide[i,4:55])), 52, 52, 52)
}

sam_fr <- NA
for(i in 1:dim(sam_wide)[1]){
  sam_fr[i] <- Spearman(1:52, as.numeric(as.vector(sam_wide[i,4:55])), 52, 52, 52)
}


caleb_fr <- NA
for(i in 1:dim(caleb_wide)[1]){
  caleb_fr[i] <- Spearman(1:52, as.numeric(as.vector(caleb_wide[i,4:55])), 52, 52, 52)
}


jordan_fr <- NA
for(i in 1:dim(jordan_wide)[1]){
  jordan_fr[i] <- Spearman(1:52, as.numeric(as.vector(jordan_wide[i,4:55])), 52, 52, 52)
}


seena_fr <- NA
for(i in 1:dim(seena_wide)[1]){
  seena_fr[i] <- Spearman(1:52, as.numeric(as.vector(seena_wide[i,4:55])), 52, 52, 52)
}
```

Checking for temporal trend within each person:

```
par(mfrow=c(5, 1), mar=c(2, 4, 1, 1))

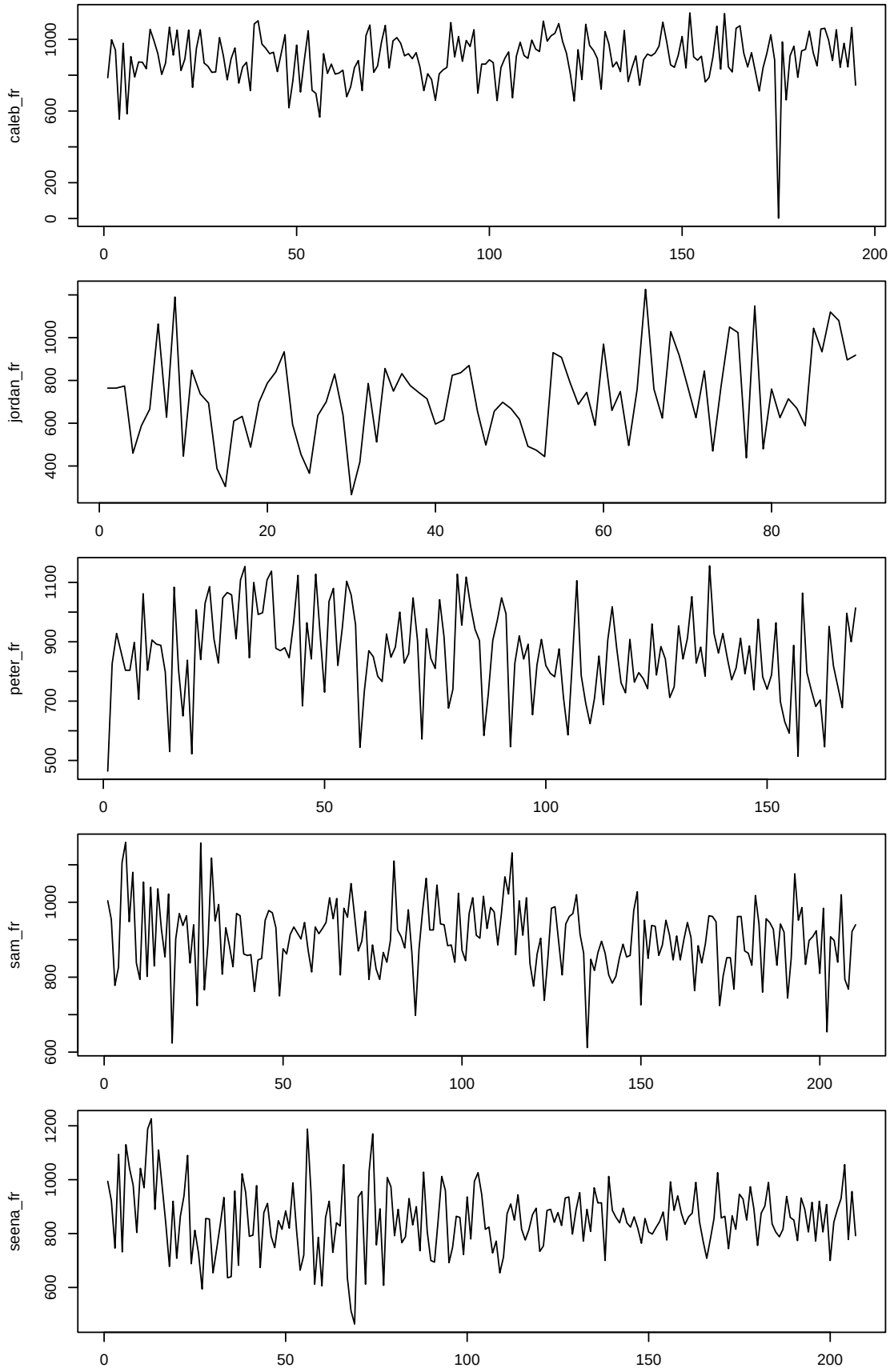
x <- 1:length(caleb_fr)
plot(caleb_fr ~ x, type='l')

x <- 1:length(jordan_fr)
plot(jordan_fr ~ x, type='l')

x <- 1:length(peter_fr)
plot(peter_fr ~ x, type='l')

x <- 1:length(sam_fr)
plot(sam_fr ~ x, type='l')

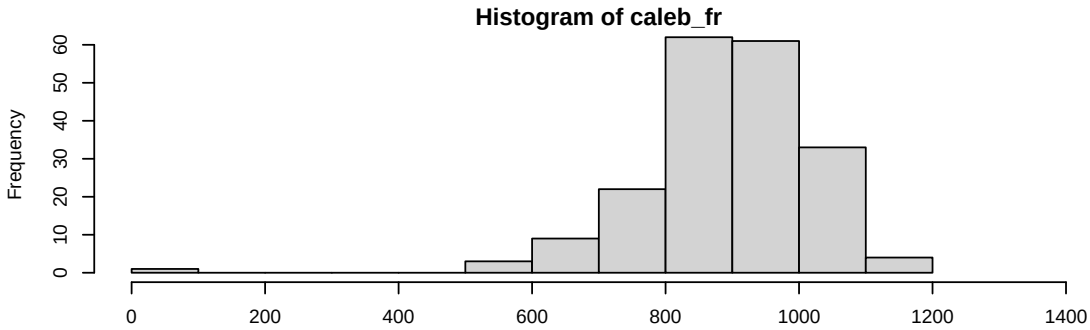
x <- 1:length(seena_fr)
plot(seena_fr ~ x, type='l')
```

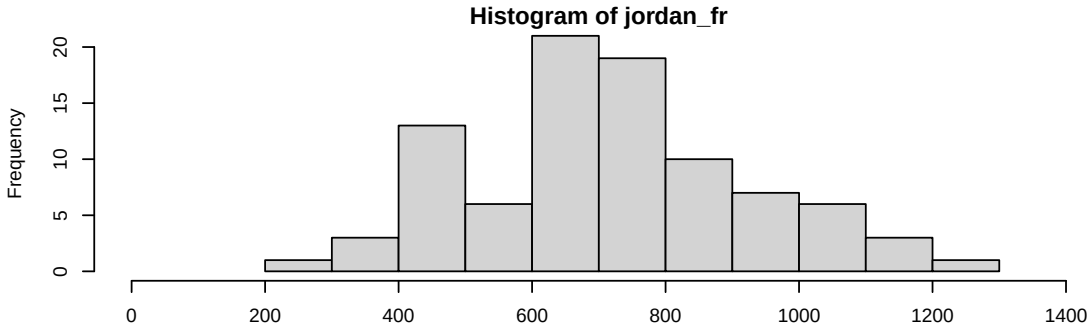

caleb_fr
0
200
600
1000
0
50
100
150
200
jordan_fr
400
600
800
1000
0
20
40
60
80
peter_fr
500
700
900
1100
0
50
100
150
sam_fr
600
800
1000
0
50
100
150
200
seena_fr
600
800
1000
1200
0
50
100
150
200

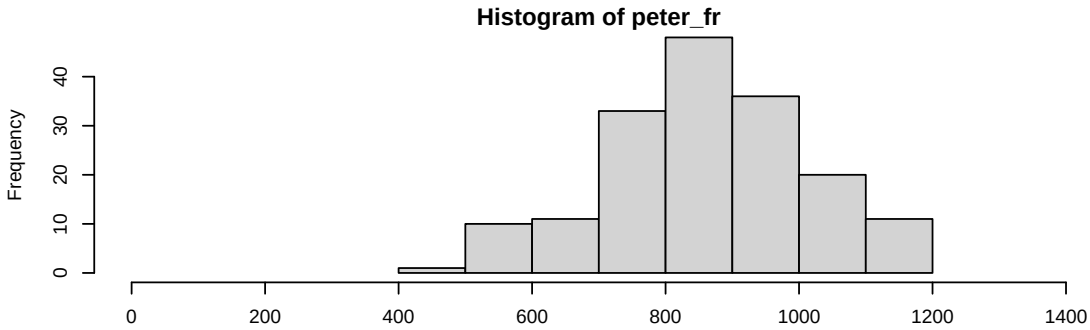
Compare distributions via histograms:

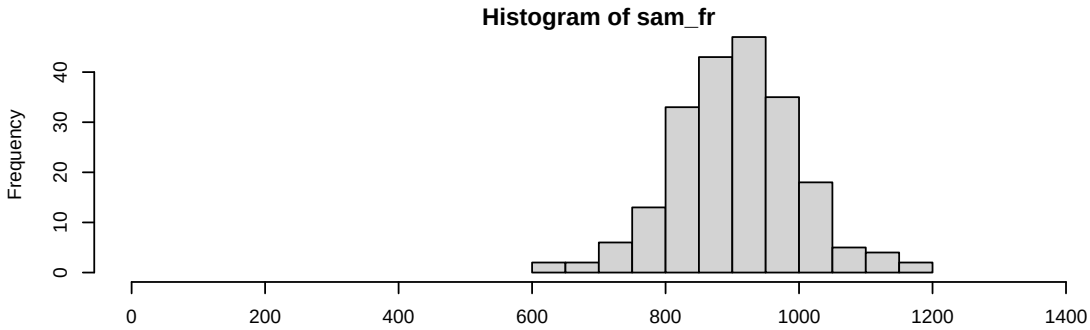
```
par(mfrow=c(5, 1), mar=c(3, 4, 1, 1))

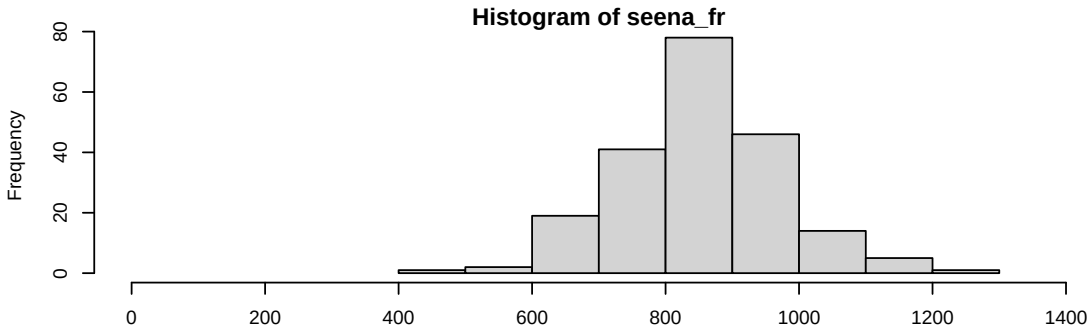
hist(caleb_fr, xlim=c(0, 1400))
hist(jordan_fr, xlim=c(0, 1400))
hist(peter_fr, xlim=c(0, 1400))
hist(sam_fr, xlim=c(0, 1400))
hist(seena_fr, xlim=c(0, 1400))
```


Histogram of caleb_fr
Frequency
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
1400


Histogram of jordan_fr
Frequency
0
5
10
15
20
0
200
400
600
800
1000
1200
1400


Histogram of peter_fr
Frequency
0
10
20
30
40
0
200
400
600
800
1000
1200
1400


Histogram of sam_fr
Frequency
0
10
20
30
40
0
200
400
600
800
1000
1200
1400


Histogram of seena_fr
Frequency
0
20
40
60
80
0
200
400
600
800
1000
1200
1400

Pairwise K-S tests:

```
ks.test(caleb_fr, jordan_fr)
```

```
## Warning in ks.test.default(caleb_fr, jordan_fr): p-value will be approximate in
## the presence of ties
```

```
##
##  Asymptotic two-sample Kolmogorov-Smirnov test
##
## data:  caleb_fr and jordan_fr
## D = 0.52051, p-value = 6.451e-15
## alternative hypothesis: two-sided
```